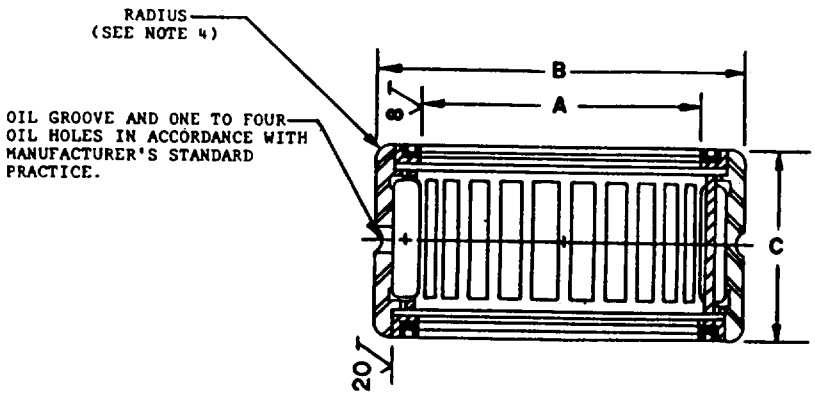
FED. SUP CLASS
3110



MS DASH NO.	A BORE DIAMETER			B OUTSIDE DIAMETER			C WIDTH +.000 -.005	RAD.	BASIC STATIC LOAD RATING LBS.	BASIC DYNAMIC LOAD RATING LBS.	APPROX. LIMITING SPEED RPM
	NOM.	MIN.	MAX.	NOM.	MIN.	MAX.					
MS52143 -1	5/8	.6258	.6267	1 1/8	1.1245	1.1250	1.000	.025	2520	2980	6100
MS " -2	3/4	.7509	.7518	1 1/4	1.2495	1.2500	1.000	.040	2540	3180	5100
MS " -3	7/8	.8759	.8768	1 3/8	1.3745	1.3750	1.000	.040	3210	3540	4400
MS " -4	1	1.0009	1.0018	1 1/2	1.4995	1.5000	1.000	.040	3670	3830	3800
MS " -5	1 1/8	1.1259	1.1268	1 5/8	1.6245	1.6250	1.250	.040	6190	5680	3400
MS " -6	1 1/4	1.2510	1.2519	1 3/4	1.7495	1.7500	1.250	.040	6530	5870	3100
MS " -7	1 3/8	1.3760	1.3769	1 7/8	1.8745	1.8750	1.250	.040	7220	6260	2800
MS " -8	1 1/2	1.5010	1.5019	2 1/16	2.0619	2.0625	1.250	.060	8130	7360	2500
MS " -9	1 5/8	1.6260	1.6269	2 3/16	2.1869	2.1875	1.250	.060	8530	7560	2400
MS " -10	1 3/4	1.7510	1.7520	2 5/16	2.3119	2.3125	1.250	.060	8940	7760	2200
MS " -11	1 7/8	1.8760	1.8770	2 7/16	2.4369	2.4375	1.250	.060	9710	8920	2000
MS " -12	2	2.0011	2.0021	2 9/16	2.5619	2.5625	1.250	.060	10030	9030	1900
MS " -13	2 1/4	2.2511	2.2521	3	2.9994	3.0000	1.750	.060	19100	14700	1700
MS " -14	2 1/2	2.5011	2.5021	3 1/4	3.2495	3.2500	1.750	.080	21600	15800	1500
MS " -15	2 3/4	2.7511	2.7521	3 1/2	3.4992	3.5000	1.750	.080	23300	16500	1400
MS " -16	3	3.0011	3.0023	3 3/4	3.7492	3.7500	1.750	.080	25800	17600	1300
MS " -17	3 1/4	3.2512	3.2524	4 1/4	4.2492	4.2500	1.750	.080	28300	25090	1200
MS " -18	3 1/2	3.5012	3.5024	4 1/2	4.4992	4.5000	2.000	.080	32200	21300	1100
MS " -19	3 3/4	3.7512	3.7524	4 3/4	4.7492	4.7500	2.000	.100	39130	31630	1000
MS " -20	4	4.0012	4.0024	5	4.9990	5.0000	2.000	.100	41930	32770	950
MS " -21	4 1/4	4.2512	4.2526	5 1/4	5.2490	5.2500	2.000	.100	44730	33960	900

MATERIAL:

A. RING - STEEL, ALLOY OR CARBON, CARBURIZING GRADE, 4620, 4720, 8620, OR 1010-1020, SAE 51100 OR SAE 52100 PER FED-STD-66 OR SAE J-404.
B. ROLLERS - STEEL, ALLOY, E50100, E51100 OR E52100 PER FED-STD-66 OR SAE J-404.
C. CAGE - STEEL, BRASS, BRONZE OR ACCORDING TO CURRENT MANUFACTURER'S STANDARD.
D. SEALS - LOW CARBON STEEL AND BUNA-N RUBBER.

HEAT TREATMENT:

A. RING - STEEL, 4620, 4720, 8620, 8720, OR 1010-1020, CASE HARDENED TO RC 58-65. CASE DEPTH .025 MIN. STEEL, SAE 51100 OR SAE 52100, THROUGH HARDENED TO RC 58-65.
B. ROLLERS - THROUGH HARDENED TO RC 58-65.

SURFACE FINISH - SURFACES MARKED $\sqrt{8}$ AND $\sqrt{20}$ SHALL HAVE MAXIMUM SURFACE ROUGHNESS IN ACCORDANCE WITH ANSI B46.1.

REVIEW ACTIVITIES: ARMY-ME, AV, MI; AF-99; DLA-IS
USER ACTIVITIES: ARMY-AR, GL; NAVY-YD, MC, SH

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APPROVED 30 APRIL 1984 REVISED

P.A. AT	TITLE BEARING, ROLLER, NEEDLE: THICK OUTER RING w/ROLLERS, CAGE AND SEALS	MILITARY STANDARD
Other Cust AF-11 NAVY-OS		MS52143
PROCUREMENT SPECIFICATION NONE	SUPERSEDES:	SHEET 1 OF 2

DD FORM 1 SEP 67 672-1 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE. 3110-0565

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PROTECTIVE COATING - UNLESS OTHERWISE SPECIFIED, PLAIN (NOT PLATED) BEARINGS TO BE COATED WITH RUST PREVENTIVE FILM BY MANUFACTURER.

LUBRICATION - BEARINGS SHALL BE FURNISHED WITHOUT LUBRICATION.

NOTES:

1. ALL DIMENSIONS ARE IN INCHES.
2. BEARINGS ARE INTENDED TO BE INSTALLED ON SHAFTS WHERE MAXIMUM TAPER DOES NOT EXCEED .0005 INCHES PER INCH OF BEARING WIDTH.
3. BEARING BORE SHOULD BE CHECKED USING A "GO" PLUG GAGE OF THE SIZE .0001 LESS THAN THE MINIMUM DIAMETER UNDER THE NEEDLE ROLLERS, AND A "NO GO" PLUG GAGE .0001 INCH LARGER THAN THE MAXIMUM DIAMETER UNDER THE NEEDLE ROLLERS. THE "NO GO" GAGE MAY ENTER 25 PERCENT OF ROLLER LENGTH.
4. ONE END OF BEARING MUST CLEAR THE MAXIMUM HOUSING FILLET RADIUS SHOWN IN "RAD" COLUMN. 1/32 MINIMUM CHAMFER ON OPPOSITE END MAY BE USED IN LIEU OF "RAD" CLEARANCE. WHEN ENDS ARE NOT IDENTICAL, THE MARKING (NOTE 8) SHALL BE APPLIED TO THE END WITH THE LARGEST SURFACE.
5. SEAL OPERATING INFORMATION - MINIMUM OPERATING TEMPERATURE MINUS 30° F AND THE MAXIMUM OPERATING TEMPERATURE SHOULD NOT EXCEED 250° F CONSTANT, 300° F INTERMITTENT. MAXIMUM PERIPHERAL SHAFT SPEED SHOULD NOT EXCEED 1000 FEET PER MINUTE. PROVISIONS FOR LEAD OR RADIUS SHOULD BE MADE ON SHAFT CORNER TO FACILITATE ASSEMBLY OF SEALED BEARINGS OF THE "RS" AND "SRS" CONFIGURATION. VENTING OF RELUBRICATION PRESSURE IS REQUIRED TO AVOID DISPLACEMENT OF SEALS IN THE "S" OR "SS" CONFIGURATION.
6. BEARINGS ARE INTENDED TO BE USED WITH HARDENED SHAFTS RC 58-65. FOR USE WITH UNHARDENED SHAFTS, BEARING SHOULD BE USED IN CONJUNCTION WITH RING, BEARING, INNER, MS51962.
7. THE MS PART NUMBER SHALL CONSIST OF THE MS DASH NUMBER, PLUS THE SEAL CONFIGURATION CODING. EXAMPLE: MS52143-2-RSS
   "S" - SINGLE SEAL, LIP INWARD
   "SS" - DOUBLE SEAL, LIPS INWARD
   "RS" - SINGLE SEAL, LIP OUTWARD
   "RSS" - DOUBLE SEAL, LIPS OUTWARD
   "SRS" - DOUBLE SEAL, ONE LIP INWARD AND ONE LIP OUTWARD.
8. THE MARKING SHALL CONSIST OF THE MS PART NUMBER AND THE MANUFACTURERS IDENTIFICATION IN ACCORDANCE WITH MIL-STD-130.
9. THE "BASIC STATIC LOAD RATING" FOR A RADIAL ROLLER BEARING IS THAT STATIC RADIAL LOAD WHICH CORRESPONDS TO A COMBINED TOTAL DEFORMATION OF ROLLER PLUS RACEWAY, AT THE MOST HEAVILY STRESSED CONTACT, OF .0001 OF THE ROLLER DIAMETER.
10. THE "BASIC DYNAMIC LOAD RATING" FOR A RADIAL ROLLER BEARING IS THAT CALCULATED, CONSTANT RADIAL LOAD WHICH A GROUP OF APPARENTLY IDENTICAL BEARINGS WITH STATIONARY OUTER RING CAN THEORETICALLY ENDURE FOR A RATING LIFE OF ONE MILLION REVOLUTIONS OF THE INNER RING. THE RATING LIFE IS DEFINED AS THE NUMBER OF REVOLUTIONS THAT 90 PERCENT OF A GROUP OF BEARINGS WILL COMPLETE OR EXCEED BEFORE THE FIRST EVIDENCE OF FATIGUE DEVELOPS, SINCE APPLIED LOADING AS GREAT AS THE BASIC DYNAMIC LOAD RATING TENDS TO CAUSE LOCAL PLASTIC DEFORMATION OF THE ROLLING SURFACES, IT IS NOT ANTICIPATED THAT SUCH HEAVY LOADING WOULD NORMALLY BE APPLIED.
11. FOR RATING LIVES AT LOADS OTHER THAN THE BASIC DYNAMIC LOAD RATINGS, AT OPERATING CONDITIONS OF INNER RING ROTATING, OUTER RING STATIONARY, STEADY LOAD, UNIFORM RPM, THOROUGH LUBRICATION, 300° F MAXIMUM BEARING TEMPERATURE, SHAFT MISALIGNMENT NOT TO EXCEED .0005 INCHES PER INCH OF BEARING WIDTH.
    EMPIRICAL FORMULA FOR LOAD-LIFE RELATIONSHIP:

$$L = \left[\frac{Cr}{P}\right]^{10/3}$$ WHERE L = RATING LIFE, MILLIONS OF REVOLUTIONS; Cr = BASIC DYNAMIC LOAD RATING, LB; P = EQUIVALENT RADIAL LOAD TO WHICH BEARING IS SUBJECTED, LB.

    FOR AN APPLIED LOAD GREATER THAN ONE-HALF THE BASIC DYNAMIC LOAD RATING, THE ABOVE LOAD LIFE RELATIONSHIP IS NOT VALID.
12. THE NEEDLE ROLLER DIAMETER VARIATION IN EACH BEARING AND THE RUN-OUT FOR EACH BEARING OUTSIDE DIAMETER AND RACEWAY INSIDE DIAMETER SHALL BE WITHIN THE LIMITATIONS PRESCRIBED IN CURRENT AFBMA STANDARDS FOR RBEC 1 TOLERANCES.
13. AVERAGE LIFE, THE LIFE WHICH 50 PERCENT OF A GROUP OF BEARINGS WILL COMPLETE BEFORE EVIDENCE OF FATIGUE DEVELOPS, IS APPROXIMATELY FIVE (5) TIMES THE RATING LIFE.
14. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
15. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS OR REQUEST FOR PROPOSAL EXCEPT REFERENCED ADOPTED INDUSTRY DOCUMENTS SHALL GIVE THE DATE OF THE ISSUE ADOPTED.
16. THE USE OF RECYCLED MATERIALS WHICH MEET THE REQUIREMENTS OF THE APPLICABLE MATERIAL SPECIFICATION WITHOUT JEOPARDIZING THE INTENDED USE OF THE ITEM SHALL BE ENCOURAGED.

APPROVED 30 APRIL 1980 REVISED

P.A. Other Cust	TITLE: BEARING, ROLLER, NEEDLE: THICK OUTER RING w/ROLLERS, CAGE AND SEALS	MILITARY STANDARD MS52143
PROCUREMENT SPECIFICATION	SUPERSEDES:	SHEET 2 OF 2

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